Competing Risks: Prepayment vs. Default

Part E(i) — Mortgage Survival Analysis

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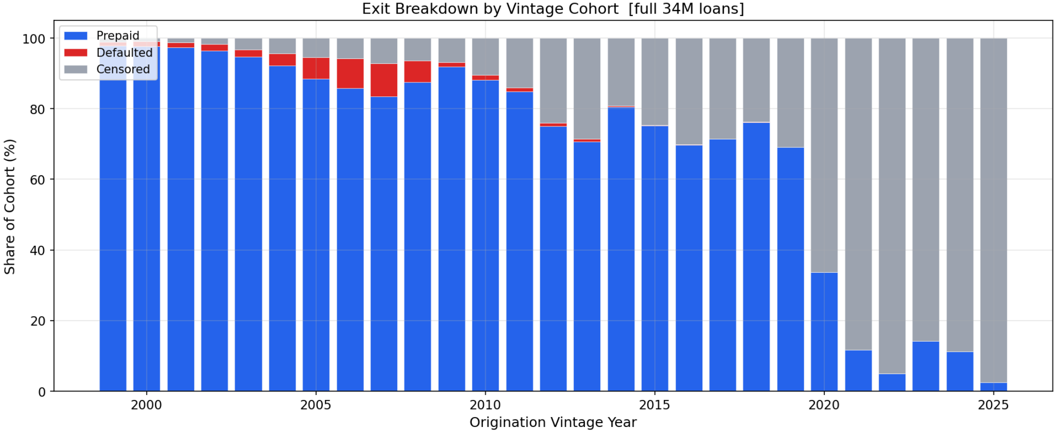
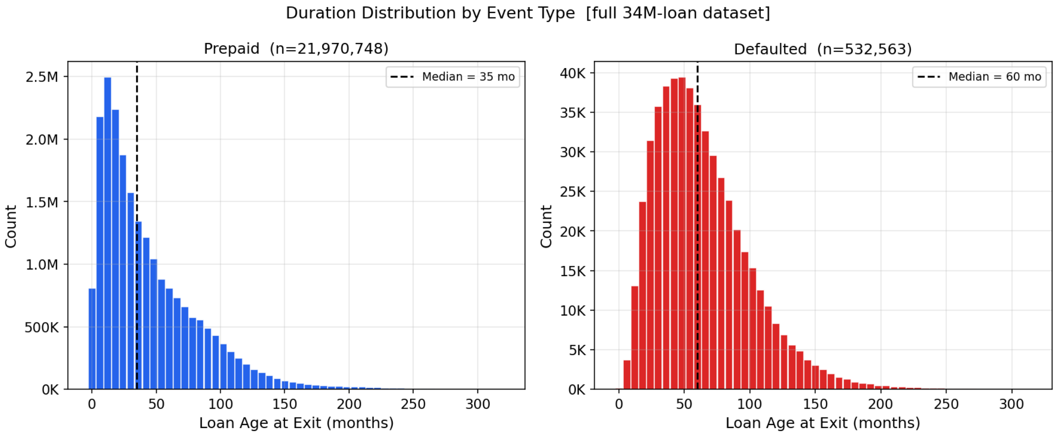
Vintage era — not FICO or LTV — is the dominant risk driver.



Two competing fates, two different borrower stories.



Two estimators, two different treatments of competing events.

Kaplan–Meier

$$\hat{F}^{(k)}_{\mathrm{KM}}(t) = 1 - \prod_{t_j \le t} \left(1 - \frac{d_j^{(k)}}{n_j^{\mathrm{KM}}}\right)$$

$d_j^{(k)}$	cause-k exits at t_j
n_j^{KM}	risk set excluding competing-event loans

Aalen–Johansen

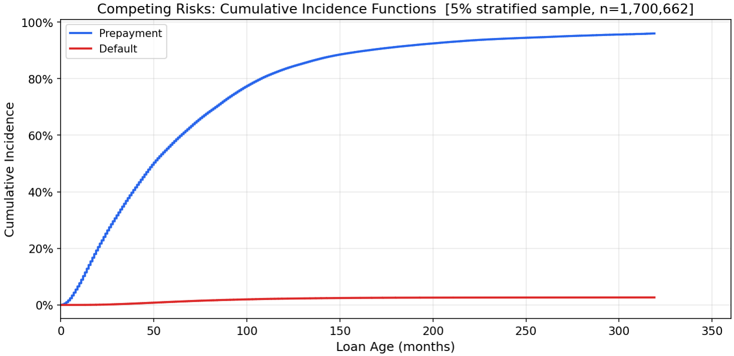
$$\hat{F}_k(t) = \sum_{t_j \le t} \hat{S}(t_j^-) \cdot \frac{d_{kj}}{n_j}$$

d_{kj}	cause-k exits at t_j
n_j	risk set including all exits
$\hat{S}(t_j^-)$	overall survival just before t_j

KM: competing exits censored $\Rightarrow$ denominator too small $\Rightarrow$ each increment inflated.

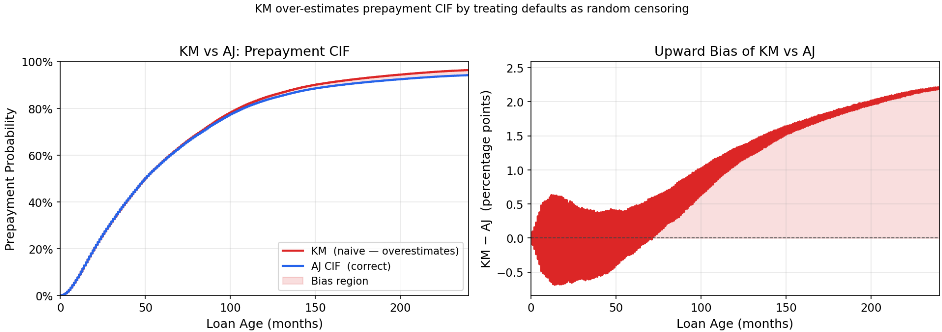
AJ: $\hat{S}(t_j^-)$ shrinks with every exit $\Rightarrow$ $\hat{F}^{(\mathrm{pre})} + \hat{F}^{(\mathrm{def})} + \hat{S} = 1$ always.

The Aalen–Johansen estimator gives the only coherent probability partition.



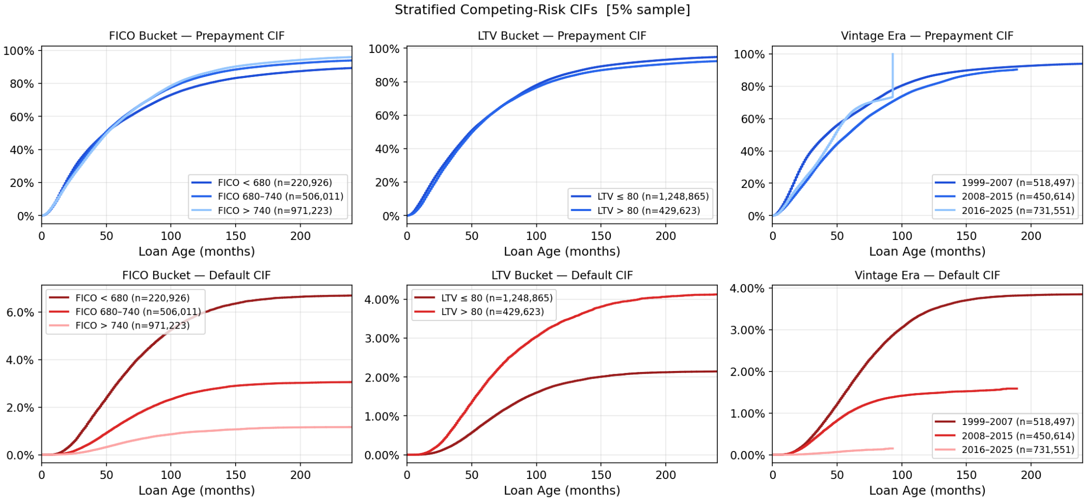
	Pre	Def	Active
5 yr	61.4%	1.1%	37.6%
10 yr	83.3%	2.2%	14.5%
20 yr	91.8%	3.1%	5.1%

KM overstates prepayment probability because it ignores competing defaults.

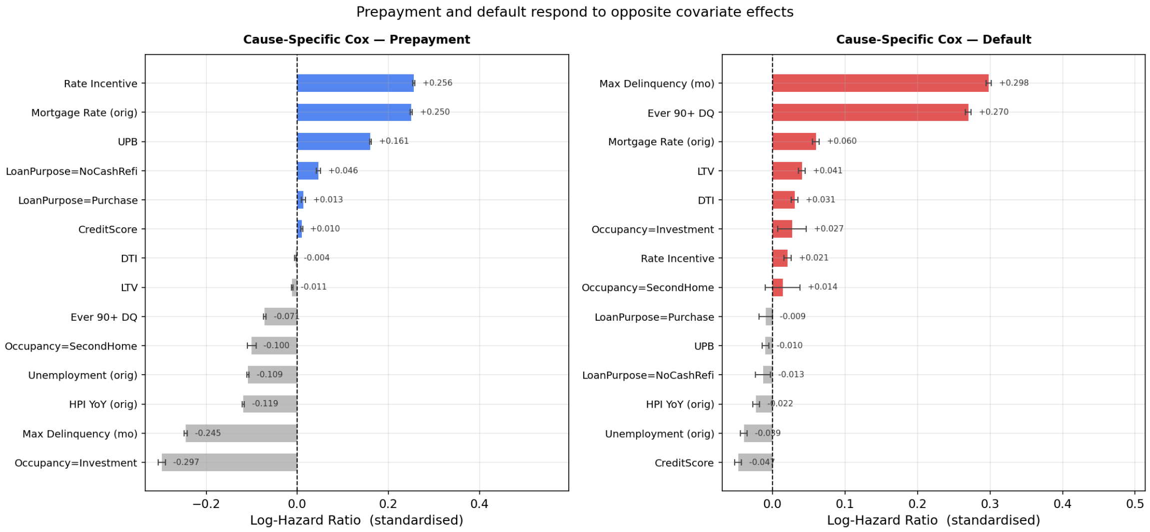


Horizon	$\hat{F}_{\mathrm{KM}}$	$\hat{F}_{\mathrm{AJ}}$	Bias
5 yr	61.9%	61.4%	+0.45 pp
10 yr	84.6%	83.3%	+1.31 pp
20 yr	93.4%	91.8%	+2.23 pp

FICO separates default 6×; LTV separates default 2× — neither separates prepayment.



Prepayment and default respond to opposite covariate effects.



Four covariates flip sign — a combined model cancels both signals.

Covariate	**Naive** $\hat{\beta}$	**Prepay** $\hat{\beta}$	**Default** $\hat{\beta}$	
Rate incentive	$+0.253$	$+0.256$	$+0.021$	(pure prepay)
UPB	$+0.161$	$+0.161$	-0.010	(pure prepay)
FICO	$+0.016$	$+0.010$	-0.047	**opposite**
LTV	-0.007	-0.011	$+0.041$	**opposite**
Max Delinquency	-0.123	-0.245	$+0.298$	**opposite**
Ever 90+ DQ	-0.035	-0.071	$+0.270$	**opposite**
Investment prop.	-0.289	-0.297	$+0.027$	diverge

Naive Max Delinquency (-0.123) hides a $+0.543$ cause-specific spread. $e^{0.256} \approx 1.29$ prepay; $e^{0.298} \approx 1.35$ default.

Key Takeaways

1. **Competing risks are not optional**: 4 covariates flip sign between causes.
2. **KM bias reaches $+$2.2 pp at 20 years**: coherent forecasting requires Aalen–Johansen.
3. **Vintage dominates**: 2007 default CIF is $6\times$ long-run average; post-2020 prepayment is frozen.
4. **FICO separates default** $6\times$**; it barely separates prepayment**.
5. **Rate incentive is the dominant prepayment signal with zero default content**: $e^{0.256} \approx 1.29$.
6. **Fine–Gray** $\approx$ **cause-specific Cox here**; the distinction is material in subprime or crisis pools.

Thank you

Questions?

Fine & Gray (1999) *JASA* · Aalen & Johansen (1978) *Scand. J. Stat.* · Sadhwani et al. (2021) *J. Fin. Econometrics*